Design patterns in PlantUML

Class-diagram parsing, rendering and controlled evolution

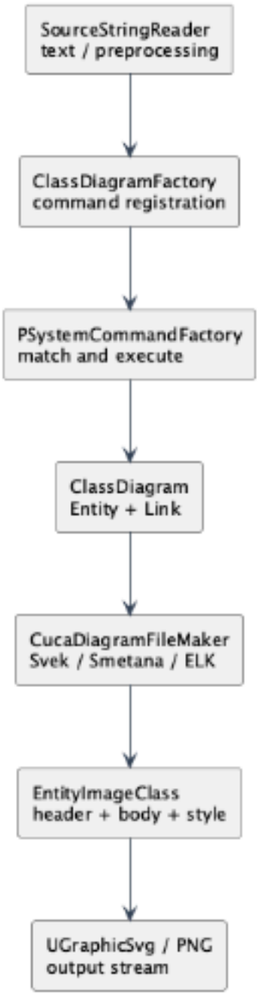
23CSE455 · Team A17 · 11 October 2026

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The path from text to SVG

```
class Order
class Customer
Order --> Customer
```

The parser selects command objects. They construct Entity and Link objects in ClassDiagram. A layout backend returns a TextBlock, which the exporter draws to SVG or PNG.



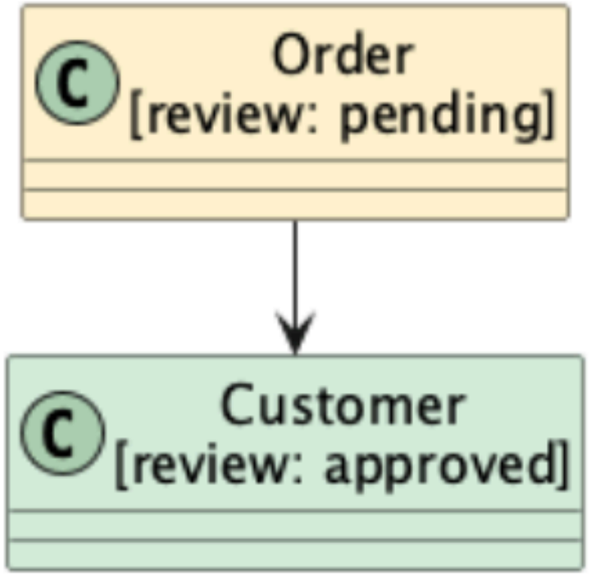
Five source-verified collaborations

Command	Class command dispatch
Factory Method	ClassDiagram product creation
Strategy	Layout backend selection
Composite	Text-block composition
Adapter	VizJs stream conversion

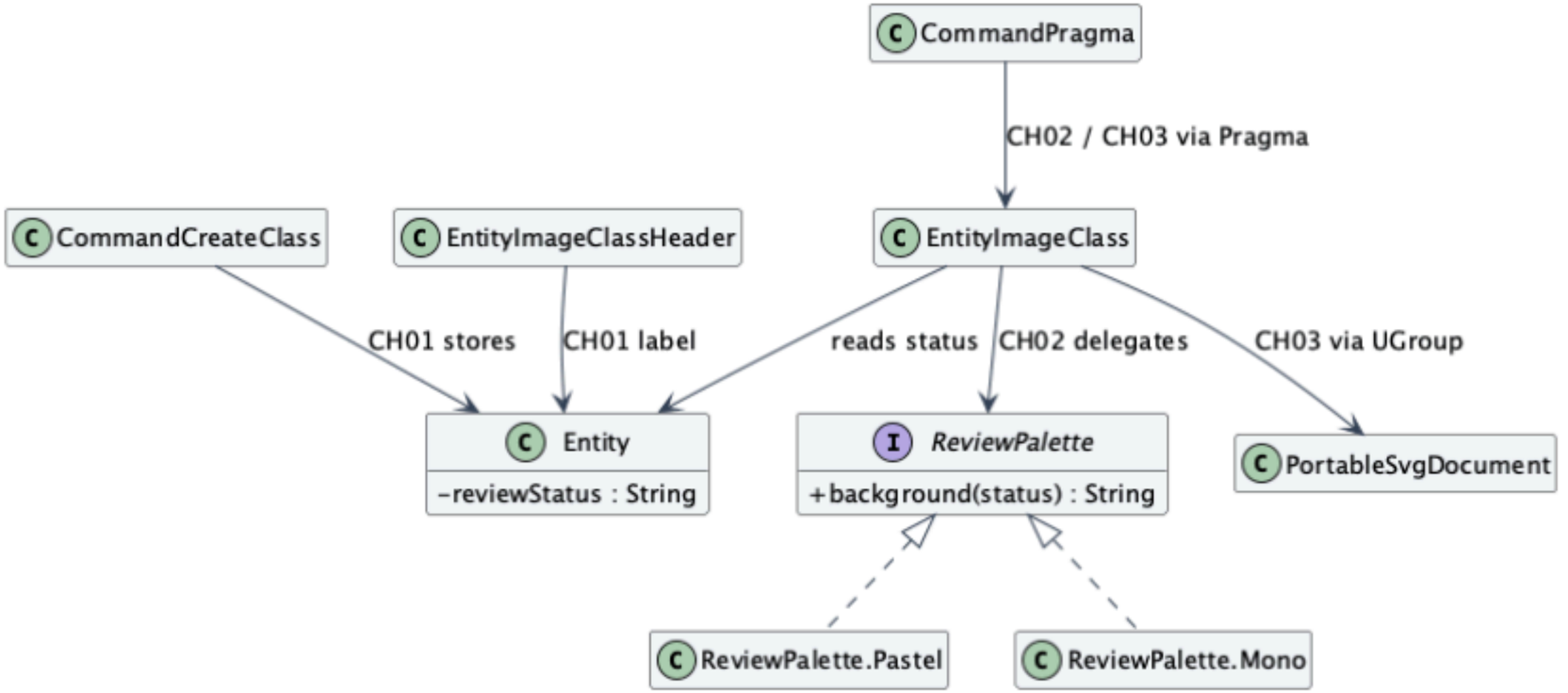
Three implemented changes

CH01 Review status
CH02 Review palette
CH03 SVG attribute

[review=pending] appears in the class header.
Pastel and mono are selectable fill policies.
Enabled SVG exports data-review-status.



The SVG serializer was a real extension point



The first metadata patch reached UGroup but produced no attribute. PortableSvgDocument accepts only named keys; adding the new key fixed the failing integration check.

Equivalent behavior in four languages

16 shared cases passed in each language

Java	106 LOC	27.206 ms
Python	101 LOC	16.827 ms
JavaScript	97 LOC	21.728 ms
C++	156 LOC	2.279 ms

Median of five fresh processes after one warm-up. Startup and input/output are included; compilation is excluded.

Verification and submission status

20 / 20 Final PlantUML integration checks passed

3,177 passed · 2 failed · 12 skipped (XML)

The console reports 10 skipped; XML includes two more skipped entries in both trees. The same Picoweb tests fail. Human review, faculty approval and repository upload remain to be recorded.

Source: pinned PlantUML commit 149874a1bb2b64c42889d7178cb89f6647ac14bc; report and evidence in A17.